
1. Internet Banking Transaction

A customer wants to transfer money.

Read:

- Account Balance
- Transfer Amount
- Account Status (**true/false**)

Requirements:

- If the account is active, check whether the balance is sufficient.
- If sufficient, display:

None

Transaction Successful

-

Otherwise, display:

None

Insufficient Balance

-

If the account is inactive, display:

None

Account Blocked

2. College Admission System

Read:

- Entrance Exam Rank
- Intermediate Percentage
- Age

Requirements:

- Students with rank ≤ 1000 and percentage $\geq 75\%$ are eligible.
- If eligible, check whether the age is **17 years or above**.
- If all conditions are satisfied, display:

None

Admission Confirmed

-

Otherwise, display appropriate messages.

3. Employee Performance Evaluation

Read:

- Performance Score (0–100)

Requirements:

Assign ratings:

- 90–100 → Outstanding
 - 80–89 → Excellent
 - 70–79 → Good
 - 60–69 → Average
 - Below 60 → Needs Improvement
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4. ATM Cash Withdrawal

Read:

- PIN
- Correct PIN
- Balance
- Withdrawal Amount

Requirements:

- Verify the PIN.
- If the PIN is correct, check whether sufficient balance is available.
- If successful, display:

None

Collect Your Cash

- Otherwise, display appropriate messages.
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5. Online Shopping Discount

Read:

- Purchase Amount
- Premium Member (`true/false`)

Requirements:

- Premium members receive:
 - 20% discount for purchases above ₹10,000
 - 10% discount for purchases above ₹5,000
 - Regular customers receive:
 - 10% discount for purchases above ₹10,000
 - 5% discount for purchases above ₹5,000
 - Otherwise, no discount.
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6. University Scholarship

Read:

- CGPA
- Attendance Percentage

- Annual Family Income

Requirements:

Scholarship is awarded only if:

- CGPA ≥ 8.5
- Attendance $\geq 85\%$
- Family income is below ₹3,00,000

Otherwise, display the reason for rejection.

7. Smart Door Security System

Read:

- Fingerprint Matched (**true/false**)
- Face Recognized (**true/false**)
- Security PIN

Requirements:

- Check the fingerprint first.
 - If matched, verify face recognition.
 - If face recognition is successful, verify the PIN.
 - Display appropriate messages for every failure.
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8. Income Tax Calculator

Read:

- Annual Income

Requirements:

Calculate the tax slab:

- Up to ₹4,00,000 → No Tax
- ₹4,00,001 – ₹8,00,000 → 5%
- ₹8,00,001 – ₹12,00,000 → 10%
- ₹12,00,001 – ₹20,00,000 → 20%
- Above ₹20,00,000 → 30%

Display the applicable tax percentage.

9. Hospital Patient Priority

Read:

- Age
- Body Temperature
- Oxygen Level

Requirements:

- If oxygen level is below 90, mark as:

None

Emergency

-

Otherwise, if temperature is above 102 and age is above 60:

None

High Priority

-

Otherwise:

None

Normal Priority

10. Campus Placement Eligibility

Read:

- CGPA
- Number of Backlogs
- Communication Score
- Aptitude Score

Requirements:

- Students with backlogs are not eligible.
 - If there are no backlogs, check the CGPA.
 - If CGPA is eligible, check communication and aptitude scores.
 - Display one of the following:
 - Eligible for Placement
 - Eligible for Written Test Only
 - Not Eligible
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